UNSOLVED OUTPUT QUESTIONS

(Increment, Decrement Operators, Java Expressions, Type Cast and Mathematical Functions)

Questions	Space provided for writing output
1. Write the corresponding Java expression / statement for the following mathematical statements: (i) $\sqrt{a^3 + b^3 + c^3}$ (ii) $3 - rs^{2r} + 4s$ (iii) $\lvert e^y - y \rvert$ (iv) $a + b / (c + d)^3$	
2. What is the output of following if x = 11 initially? (i) 10 * ++ x – 1; (ii) ++ x + x ++;	
3. What is the output of following expression if x = 2 initially? x += x + + + + + x / 2;	
4. What will be the output of following if x=5 ? (i) + + x * 2; (ii) x + + * 2;	
5. If y = 8 and x = 13 then output the value of **'c'**. c = + + y + x + + + y;	
6. What will be the value in variable **'f'** after executing the following expression if a = 5 and b = 3: f = (+ + b) * b + + - - - a;	
7. If x = 5 initially, What will be the value of **'y'** after execution of : (i) y = (5* + + x) % 6; (ii) y = x * + + x;	
8. If x = 3 initially, what will be the result **'x'** after the execution of following statement: x + = x + + + - - x + 4;	
9. If x = 6 initially what will be the result/output of the following two expressions? (i) 10 * + + x; (ii) x + + * 10;	
10. What will be the value of **'x'** after execution of following statements: (i) x = ('B' + 10) * 3; (ii) x = ('A' + 'c') + 5;	
11. What is the output of expression **(float)1/3** ?	
12. What is the value of **'x'** after evaluating x % = x + + + + + x – 2; if **x = 3** initially ?	

13. What will be the output of following relations if **d = 20** initially?
(i) + + d<=20 (ii) d + + <=20
(iii) + + d<= 35 (iv) (+ + d) *10>d

14. What is the output of following if **a = 2** initially ?
a + = a + + + + + a / 2;

15. What is the output of following if **y = 11** initially?
(i) 10 * + + y – 1;
(ii) + + y + y + +;

16. If **x = 10** and **y = 20**. Find the value of '**Z**'.
Z = x + + + + + y – x;

17. Give an output for following code:
```
int m = 6, n = 9;
System.out.println("M + N = " + m + n );
```

18. If **m = 5** and **n = 3** output the values of '**m**' and '**n**' after execution in (i) and (ii)
(i) m - = n;
(ii) n = m + m / n; **ICSE 2005**

19. Write valid Java statements for the following.
(i) To store the square root of **num**.
(ii) To print the **3** raised to the power **4**.
(iii) To print two integers **x, y** in the same line.
(v) To print the square root of **-16** after converting it into positive using mathematical functions only.

20. Give the output of following program code:
```
class test
{
int v;
v = (int) Math.sqrt(Math.abs(-16));
System.out.println("The result = "+v);
}
```

21. What is the value of :
(Math.abs(-1) + Math.pow(2, 3)); ?

22. If **z = 10**, find the value of '**M**' for the expression:
int M = (5 * + + z) % 4;

23. Name the process and output which is being generated by the function of class?
```
class test
{
public void display( ) throws IOException
{
double a = Math.sqrt( Math.abs(-49));
System.out.println("The Result = " + a );
} }
```

Output O

24. Find the values of **'a'** and **'b'** after following functions get executed: **(i)** a = Math.ceil(38.76); **(ii)** b = Math.floor(22.76);	
25. Explain the following **Math** functions: **(i)** Math.ceil(x) **(ii)** Math.floor(x)	
26. If **m = 5** and **n = 2** what will be the output of following two statements? **(i)** m = m++; System.out.print(m); **(ii)** n = n / m; System.out.print(n);	

27. Give the output of following program code:
public static void main( )
{
char c = 'A';
short m = 26;
int n = c + m;
System.out.print("n = " + n);
}	

28. What is output of following?
char c = 'a';
short n = 25;
int m = c + n;
System.out.println(m);	

29. What will be output of following? int y = 10; System.out.println(++ y); System.out.println(y ++);	
30. What will be the output of following code: int a = 10, b = 20; System.out.println(a + b+ " is equal to "+ (a + b));	

31. Find and correct the errors in the following program code:
int x = 7; v = 10;
x + y = q;
System.out.println(q)	

32. Give an output for following code: int x = 5, y = 7, B; B = x++ +x++- --y; System.out.println("B = " + B);	

33. What is output of following program code? char a = 'x'; int b = 2; int c = 4; char d = 'y'; System.out.println(a + d); System.out.println(b + c); System.out.println(a + b); System.out.println(c + d);	
34. What will be the value of **'x'** after evaluating if **x = 5** initially: x *= + + x + -- x + 5;	

35. What will be output of the following code?
int m = 0, n = 30, q = 40;
m = - - n + q + + + n;
System.out.println("M = " + m);	

36. Give the output of following program segment : System.out.println("Result 1 = " + 7 + 5); System.out.println("Result 2 = " + (7+5));	

37. What will be the output of the following code ?
int m= 605/10 + 45%8 + 29%13;
char ch = (char) m;
System.out.print(ch);	

38. Give the output for following program code if **x = 7** and **y =11**: System.out.println("x != y " + (x != y)); System.out.println("x == y " + (x == y));	
39. What will the following statement print ? System.out.println(Math.max(9,Math.min(13,12)));	
40. What is the result stored in **'x'** after evaluating the following : int x = 5; x = x + + * 2 + 3 * - - x; **ICSE 2010**	
41. What will the given functions returns when invoked: (i) Math.max(-17, -19) (ii) Math.ceil(7.8) **ICSE 2010**	
42. Give output of following program: class testOutput { public static void show( ) { double a = 6.0, b = 2.0; System.out.println("a + b = " + (a + b)); System.out.println("a - b = " + (a - b)); System.out.println("a * b = " + (a * b)); System.out.println("a / b = " + (a / b)); } }	

43. Give output of following program:
class testOutput
{
public static void show( )
{ char ch1 = 'A', ch2 = 'B', ch3 = 'b';
int a = ch1; int b= ch2; int c = ch3;
System.out.println("ch1="+ ch1 + "\t" + "a="+ a);
System.out.println("ch2="+ ch2 + "\t" + "b="+ b);
System.out.println("ch3="+ ch3 + "\t" + "c="+ c);
} }	

44. What will be the output of the following code ?
int k = 5, j = 9;
k += k++ - ++j + k;
System.out.println("k = " + k);
System.out.println("j = " + j); ICSE 2011	

45. What will be the output of the following code ?
double b = -15.6;
double a = Math.rint(Math.abs(b));
System.out.println("a = "+ a); ICSE 2011	

46. Write the Java expression for $\sqrt{2as+u^2}$ ICSE 2012	
47. State the values of ***n*** and ***ch***. ICSE 2012 char c = 'A'; int n = c + 1; char ch = (char) n;	
48. What will be the result in ***x*** after evaluating the following expression? ICSE 2012 int x = 4; x += (x++) + (++x) + x;	
49. Give the output of the following code: ICSE 2012 double x = 2.9, y = 2.5; System.out.println(Math.min(Math.floor(x), y)); System.out.println(Math.max(Math.ceil(x), y));	
50. What are the *type of casting* shown by the following example? ICSE 2013 double x = 15.2; int y = (int) x;	
51. Write a Java expression for $ut + \frac{1}{2}ft^2$ ICSE 2013	

52. What are the final values stored in variables **x** and **y** below? ICSE 2014
double a = – 6.35; double b = 14.74;
double x = Math.abs(Math.ceil(a));
double y = Math.rint(Math.max(a, b));	

53. Name the mathematical function which is used to find **sine** of an angle given in radians. ICSE 2015	
54. Write the Java expression for : ICSE 2015 **$(a^2 + b^{2)}$ / 2ab**	

55. If **int y = 10**, then find : **ICSE 2015** int z = (y++ * (y++ + 5));	
56. Give the output of the following ***Math*** functions: **ICSE 2016** **(i) Math.ceil(4.2)** **(ii) Math.abs(-4)**	
57. Write the Java expression for: $\mathbf{T} = \sqrt{A^2 + B^2 + C^2}$ **ICSE 2016**	
58. Give the output of the following expression : **ICSE 2016** **a += a++ + ++a + --a + a--;** **when a=7**	
59. Write the Java expression for : $\mathbf{ax^5 + bx^3 + c}$ **ICSE 2017**	
60. State the final value if ***x1***, if ***x = 5***? **ICSE 2017** **x1 = ++x - x++ + - -x;**	
61. What are the values stored in **r1** and **r2**: **ICSE 2017** (i) double **r1** = Math.abs(Math.min(-2.83, -5.83)); (ii) double **r2** = Math.sqrt(Math.floor(16.3));	

if, else-if, if-else-if, Nested if, switch-case statements and *Ternary Operators*

Questions	Space provided for writing output

1. Give output for following program code:
int weight = 70;
   System.out.print("You are");
   if(weight > 60)
        System.out.println(" over weight");
   else
        System.out.println(" under weight");	

2. Give output of following code:
int s = 4;
   if (s >10)
        System.out.println("Good");
   else
        System.out.print("Better");
   System.out.print(" Best");	

3. What will be output of following code ?
int a = 10, b = 20;
if((a < b) || (a = = 5))
   System.out.println(a);
else
   System.out.println(b);	

4. Give output of following program code:
`int x, y;`
`x = 15; y = 10;`
`if(x > y)`
`{`
`  if (y > 5)`
`   System.out.println("Y = " + y);`
`}`
`else     System.out.println("X = " + x);`	

5. Rewrite the following using **Ternary Operator** :
`int age = 20;`
`boolean res;`
`  if(age >= 18)`
`        res = true;`
`  else`
`        res = false;`	

6. Find the value of '**K**' after execution of the following statement: `K = 1000 + 500 > 1500 ? 25 : 20;`	

7. Rewrite the following codes using only **ternary/ conditional operators**:
(i) `if( x > 5)`
`   int y = x+2;`
`else int y = x -2;`	

(ii) `if (sales > 25000)`
`        double amt = sales*20.0/100.0;`
`else`
`        double amt = sales*10.0/100.0;`	

(iii) `if(marks > 200)`
`char res = 'P';`
`else char res = 'F';`	

8. Underline **errors** and rewrite following codes after correcting the errors:
(i) `if( x => 25)`
`   int x + 2 = y;`
`else int y = x -2;`	

(ii) `if (sales > 25000 & < 500000)`
`double amt = sales*20.0%;`
`else double amt = sales*10%`	

(iii) `if(marks > 200)`
`char res = 'P' ;`
`else res = 'F'`	

9. Underline **errors** and write the correct code for following:

```
Int c, m
If( c = 3 );
    m = 500;
else
    m = =200;
```

10. Write **true/false** for the following expressions:
(i) 30 <= 30
(ii) -20 > -5
(iii) 10 < 10
(iv) 24 >= 34
(v) -6 > -32

11. Write **true/false** for the following expressions:
(i) 19 == 19
(ii) 1+7 = = 9
(iii) 9-3 > 5
(iv) 2+5*3 >= 12
(v) -6-10 > 32

12. Write the given code using *'if'* statement to avoid number of comparisons :

```
if( s == 1)
    System.out.println("One");
if( s == 2)
    System.out.println("Two");
if( s == 3)
    System.out.println("Three");
if( s == 4)
    System.out.println("Four");
if( s == 5)
    System.out.println("Five");
```

13. Rewrite the following code using **switch-case** :

```
char ch = 'A';
if( ch == 'A' )
    System.out.println( "A" );
else if( ch == 'Z' || ch == 'z' )
    System.out.println("Z");
else
    System.out.println("Invalid");
```

14. Give output of following code when the function **print( )** is invoked with the value of **choice** is **'a'**, **'c'** and **'n'** :

```
class test
{
public static void print( char choice )
```

```
{
switch( choice )
{
case 'a':   System.out.print("Java");
case 'b':   System.out.print(" is");
case 'c':   System.out.print(" programming");
case 'd':   System.out.println(" language");
default :  System.out.println("Simple program");
} } }
```

15. Give the output of given code for value of **'st'** is:

(i) 'A' (ii) 'C'
(iii) 'D' (iv) 'F'

```
switch( st )
{
case 'A':  System.out.print("Grade A");
case 'B':  System.out.print("Grade B");
case 'C':  System.out.print("Grade C");  break;
case 'D':  System.out.println("Grade D");
default :  System.out.println("Grade F");
}
```

16. Find **error** and **rewrite** the correct form for following program code:

```
switch( ch );
{
case 'a':   n = n + 2; t = t + 3
            System.out.print( n + " " + t ); break;
case "b":   n = n + 4;  t = t + 7;
            break
            System.out.print( n + " " + t )
}
```

17. Write alternative coding using **switch-case** for following program code:

```
if( s == 1)
    System.out.println("One");
if( s == 2)
    System.out.println("Two");
if( s == 3)
    System.out.println("Three");
if( s == 4)
    System.out.println("Four");
if( s == 5)
    System.out.println("Five");
```

18. Write the alternative coding using **switch-case** for following program code :

```
if( s == 'A' || s == 'a')
System.out.println("Vowel A");
if( s == 'E'  || s == 'e')
System.out.println("Vowel E");
if( s == 'I' || s == 'i')
System.out.println("Vowel I");
if( s == 'O'  ||  s == 'o')
System.out.println("Vowel O");
if( s == 'U'  ||  s == 'u')
System.out.println("Vowel U");
```

19. Consider the following program code and Give output for each value of **'ch'** :

(i) ch = 'A'; (ii) ch = 'o' (iii) ch = 'i'

(iv) ch = 'e' (v) ch = 'u' (vi) ch = 'f'

```
int  x = 0;
switch( ch )
{
case  'a':    x + = 1; break;
case  'e':    x + = 1;
case  'i':    x + = 1; break;
case  'o':    x + = 1;  System.out.print( x + "  "); break;
case  'u':    x + = 1; System.out.print( x );
default : System.out.print("Not a lowercase vowel");
}
System.out.println("It is a switch based program");
```

20. Give the output of the following program code for **res = 1** and **res = 2**.

```
int  res, K = 6;
if (res == 1)
            System.out.println  (K++);
else if (res >= 2)
{
            System.out.println  (++K);        K++;
            System.out.println  (++K * 3);
}
```

21. What will be the output of following program code when:

(i) val = 250 and (ii) val = 1600

```
    int  val, amt, n = 700;
amt = n + val > 1800 ? 2100 : 1100;
System.out.println( amt );
```

Output Questions

ons

22. What is the value of variable **ans** after evaluating the following statement if **ans = 200** initially: ans = (ans >= 180) **?** ans++ **:** ++ans;	

23. What will be output of following code if :
(i) pr = 2000 **(ii)** pr = 500 **(iii)** pr = 1000
int v, out, num=1000;
out = num + pr > 1850 **?** 700 **:** 400;
System.out.println(out);	

24. Rewrite the given code using **Ternary operator** :
if (num > 0) res = 1;
else res = 2;	

25. The following is a segment of a program. **ICSE 2009**
x = 1; y = 1;
if(n > 0)
{
x = x + 1; y = y – 1;
}
What will be value of **x** and **y**, if **n** assumes a value
(i) 1 **(ii)** 0 ?	

26. Give the output of the following code when :
(i) opn = 'b' **(ii)** opn = 'x' **(iii)** opn = 'a' **ICSE 2009**
switch(opn)
{
case 'a': System.out.println ("PlatformIndependent");
break;
case 'b': System.out.println ("ObjectOriented");
break;
case 'c': System.out.println ("Robust and Secure");
break;
default : System.out.println ("Wrong Input");
}	

27. Give the output for the following : **ICSE 2010** char x = 'A'; int m; m = (x == 'a') ? 'A' : 'a'; System.out.println("m = " + m);	

28. What are the values of variables ***x*** and ***y*** when the given statements are executed? **ICSE 2012**
int a = 63, b = 36;
boolean x = (a > b) ? true : false;
int y = (a < b) ? a : b;	

29. Rewrite the following program segment using the ***if..else*** statement : **ICSE 2013** comm = (sale > 15000) ? sale*5/100 : 0 ;	

30. Rewrite the following program segment using the ***if..else*** statement : **ICSE 2014** String grade = (mark>=90) ? "A" : (mark>=80) ? "B" : "C";	
31. Evaluate the value of **'n'** if, **p =5, q =19** : **ICSE 2015** int n = (q - p) > (p - q) **?** (q - p) : (p - q);	

32. Rewrite using ***ternary operator*** : **ICSE 2016**
if(x % 2 == 0)
System.out.print("EVEN");
else
System.out.print("ODD");	

ITERATION STATEMENTS

(for( ), while( ), do-while( ) and their nested forms)

Questions	Space provided for writing output

1. Give output of the following Java code:
for(int i = 10; i >= 1; i --)
{
System.out.print(i + ",");
}
System.out.println(i);
	int i = 10
	while(i>=1)
	{ System.out.print(i+", ");
	i--;
	} System.out.println(i);

2. What will be output of the following code: int y; for(y = 0; y <= 5; ++y) System.out.println(y); System.out.println(y);	0 1 2 3 4 5 6

3. What is the output of the following program code:
int y =4;
for(; y <= 16; y += 2)
System.out.println(y);
System.out.println(y);	

4. Correct errors (*if any*) in the following program code and then write the output:
int y = 3;
for(; y <= 13, y += 4)
System.out.println(y),
System.out.println(y);	

5. What will be the output of following code: int y; for(y = 1; y <= 10; y += 2) System.out.print(y +" "); System.out.println("\n"+ y*10);	
6. Write the output of following program code: int a; for(a = 10; a >= 0; a --) System.out.print(a + " "); System.out.println(a);	

7. What will be the output of following program code:
```
int y = 5;
for( ; y <= 19; y  += 2 )
    System.out.println( y );
```

8. Write a valid Java statement to create a reverse loop from 8 to 0.

9. What will be the output of following code :
```
int a;          for( a = 10; a >= 0;  -- a )
                {
                }
                System.out.println( a );
```

10. What will be the output of following code :
```
int  f = 1, i = 2;
do
{  f *= i ;
} while( ++ i < 5);
System.out.println( f );
```

11. Give the output for following program code:
```
class  check
{
public  void  dataValue( )
{
int i =3, h = i;
do
{
   i++;
} while( i < 0 );
System.out.println( h + "\t" + i );
} }
```

12. Give the output of following Java code:
```
int  i = 0;
for( i = 10; i >= 1;  i -- )
{
   if( i % 2 == 1 )
            continue;
System.out.print( i + "," );
}
```

13. State the output of following function when invoked :
```
public void Display( )
{
for( int i = 0; i < 4;  i++ )
{
   for( int j = 1 ; j <= 3 ; j++ )
   {
         if( j == 2 )
              continue;
```

System.out.println(i + "\t" + j); } } }	
14. Write valid Java statement(s) to print **float** numbers from 70 to 51 from the steps 0.5.	
15. Using **while( )** function and suitable variables, print the integers from 21 to 51.	

16. State the output of the following function :
 public void show( )
 {
 long y = 838383, n = y, p, s;
 while(n > 0)
 {
 p = n % 10;
 System.out.println(Math.pow(p, 2));
 n = n / 100;
 }
 System.out.println(y);
 }	

17. State be the output of following program code :
 long num = 37373737, t = num, q = 0;
 System.out.println(num);
 while(t > 0)
 {
 q = q + t % 10 ;
 t /= 100;
 }
 System.out.println("Final Output = " + q);	

18. What will be the output of following code?
 long num = 53535353, t = num;
 System.out.println(num);
 while(t > 0)
 {
 System.out.println(t % 10);
 t /= 100; }	

19. Give the output of the following function :
 public static void main(String args[ ])
 { int i = - 8;
 do {
 System.out.println(i);
 i ++ ;
 }while(i < 0);
 }	

20. Give the output of following program code:

```
public class result
{
void  Afunction( )
{    int x = 15;
while( x >= 0)
{    x -- ;
System.out.println( x );
if ( x == 10)
break;
}   }  }
```

21. Answer the questions for the following code :

```
int Afunction( int n )
{
int g = 1, x = 1;
while( x <= n )
{
   g *= x;
   ++x;
}
return ( g );
}
```

(i) What will be returned by the function if **n=4**?

(ii) What will be returned by the function if **n=7**?

(iii) State, what is being computed by the function?

22. Rewrite the given program code using **while( )** :

```
int i = 0;
for(i = 2 ; i <= 20; i += 2 )
   System.out.println( i + ",");
```

23. What will be the output of the given function when invoked?

```
public void show( )
{
long num = 234455, sum = 0;
do
{
   sum *= 10;
   long y = num % 10;
   sum += y;
   num /= 10;
}while(num != 0);
System.out.println("The Output = " + sum);
}
```

24. Write the equivalent **do-while** loop for the following program code:

```
int n = 40;
for( int y =10; y >= 2; y -- )
   n ++;
```

25. Give the output of the given function when invoked?
public void Print( )
{
char ans = 'A'; int x = 0;
do
{
x++;   System.out.print(x + " " + ans + " ");
ans += 2;
}while(x < 7);
System.out.println("Loop Terminates"); }	

26. Analyze the following program segment and determine how many times the body of loop will be executed (show the working). **ICSE 2009**
int x = 5, y = 50;
while (x <= y)
{ y = y / x;
System.out.println(y);
}	

27. Convert the following segment into an equivalent **do-while** loop : **ICSE 2009**
int x, c;
for(x = 10, c = 20; c > = 10; c = c – 2)
x++;	

28. Convert the following segment into **for( )** loop.
int x = 2, y = 8;
do
{
x ++;
y += 2;
System.out.println (x + "," + y);
}
While (x < 20);	

29. State that how many times the following loop runs (show working) : for (int y = 3; y <= 30; y += 3)	

30. What will be output of the following code?
int m = 2, n = 15;
for (int i = 1; i < 5; i++) ;
m++; -- n;
System.out.println("m = " + m);
System.out.println("n = " + n); **ICSE 2010**	

31. Analyse that how many times the following loop executes and what will be the output ?
int k = 1, i = 2;
while (++ i < 6)
k *= i;
System.out.println(k); **ICSE 2010**	

32. Analyze that how many times the following loop executes and what will be the output ?

```
int p = 200;
while ( true )
{
    if( p < 100 )     break;
p = p - 20;
}
System.out.println(p);
```

ICSE 2011

33. Rewrite the following program segment using ***while( )*** instead of ***for( )*** statement.

```
int f = 1, i ;
for( i = 1; i <= 5; i++ )
{
f *= i;   System.out.println( f );
}
```

ICSE 2012

34. How many times will the following loop execute? What value will be returned?

```
int x = 2, y = 50;
do
{
        ++x;
        y -= x++;
} while( x <= 10 );
return y;
```

ICSE 2013

35. What is the final value of ***ctr*** after the iteration process given below, executes?

```
int ctr = 0;
for(int i = 1; i <= 5; i++ )
    for(int j = 1; j <= 5; j += 2 )
        ++ctr;
```

ICSE 2013

36. Study the method and answer the given questions.

```
public void sampleMethod( )
{
 for( int i = 0; i < 3;  i++)
 {
for( int j=0; j<2; j++)
{
int number = (int)(Math.random( ) * 10);
System.out.println(number);
} } }
```

(i) How many times does the loop execute?

(ii) What is the range of possible values stored in the variable *number*?

ICSE 2014

37. Write the output of the following code :

```
char ch;  int x = 97;
do
{
        ch = (char) x;
        System.out.print( ch + " " );
        if( x % 10 == 0 )  break;
    ++x;
}while( x <= 100 );
```

ICSE 2015

38. Convert the following ***while( )*** loop to its corresponding ***for( )*** loop:

```
int m = 5, n = 10;
while( n >= 1 )
{
System.out.print( m*n ); n--;
}
```

ICSE 2016

39. Analyze the given program segment and answer the following questions:

(i) Write the output of the program segment.

(ii) How many times does the body of the loop gets executed?

```
for( int m = 5; m <= 20;  m += 5 )
{     if( m % 3 == 0 )
          break;
      else
        if( m % 5 == 0 )
          System.out.println( m );
            continue;
}
```

ICSE 2016

}

SINGLE DIMENSIONAL ARRAY and DOUBLE DIMENSIONAL ARRAY

Questions	Space provided for writing output
1. What is *wrong* in the following code ? Explain. Also write correct form of the following: int u[ ] = new int[6]; u = { 2, 5, 6, 7, 8, 9 };	
2. Is any of the following is incorrect? Explain. **(i)** int y[ ] = new int[100]; **(ii)** int [ ]d = new int[100];	
3. Given that: int b[ ] = { 2, 66, 76, 23, 78, 96 }; Give the subscripts of 76 and 78.	

Output Questions

4. What will be output of the following code? int x[ ] = { 4, 8, 2, 6 }; int y[ ] = { 6, 9, 4, 8 }; x = y; System.out.print("\n x[2] = " + x[2]); System.out.print("\n y[2] = " + y[2]);	
5. What will be output of the following code? int [ ] ar = { 2, 5, 8, 9, 1, 6, 4, 9, 3, 5 }; System.out.println(ar.length);	
6. What will be the output of following code? int ar[ ] = { 2, 5, 8, 9, 1, 6, 4, 9, 3, 5 }; System.out.println(ar [3+4]); System.out.println(ar[1+3]); System.out.println(ar[5-3]);	
7. Find the errors and correct them. (i) int ar = { 2, 5, 7, 9, 4, 6 }; (ii) int m[16] = { 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31 };	
8. Write a valid Java statement to declare and also initialize a character array of uppercase vowels.	
9. Given that d[ ] = {4, 5, 6, 1, 2, 3}; What will be the values of **d[0+2]** and **d[2-2]**?	
10. Consider an array **mx[ ]** declared and initialized with *float* type elements. Write a valid java statement to show that, how many elements are in the array?	
11. What will be the output of following code? int ar[ ] = { 2, 5, 8, 9, 1, 6, 4, 9, 3, 5 }; System.out.println(ar [3] * 2); System.out.println(ar[4] + 3); System.out.println(ar[6+2] * 3);	
12. What will be the output of following code? int ar[ ]={ 2, 5, 8, 9, 1, 6, 4, 9, 3, 5 }; (i) State the start and end indices of **ar[ ]**. (ii) State the element at 3rd and 7th index. (iii) Find and print the middle index and the element present. (iv) State whether sum of elements from 0 to 4 indexes are more than sum of elements from 5 to 9 index or not.	
13. Write Java statements to perform the following: (i) To declare a single subscripted variable of 20 integers.	

(ii) To declare single subscripted variable of 14 real numbers. (iii) To initialize first 10 even integers in a single dimensional array.	

14. Give the output of the following function :
`void show( )`
`{`
`int y[ ] = {2, 4, 5, 8};      int q = y.length; int p=0;`
`for(int j = 0; j < q; j++)`
`{`
`p = y[ j ] + y[ 3 – j ];      System.out.println (p);`
`} }`	

15. Give the output of the following code :
`int ar[ ] = {3, 4, 5, 6};`
`for(int j = 0; j < 4; j++)`
`{`
`for(int n = 0; n <= j; n++)`
`{`
`System.out.print(ar[ n ]+ " ");`
`}  System.out.print("\n");`
`}`	

16. Give the output of the following code :
`int z[ ] = {39, 42, 36, 45, 75, 98};`
`System.out.print(z[ 3 ] + "," + z[ 4 ] * 2);`	

17. Give the output for the given code :
`int num = 96258, n = 0;      int b[ ] = new int[5];`
`while(num > 0)`
`{`
`b[ n++ ] = num % 10;      num /= 10;`
`System.out.println(b[ n-1 ]);`
`}`	

18. What will be the values stored in variables ***a*** & ***b*** after the given statements gets executed ?
`int arr[ ] = {6, 4, 8, 5, 9};`
`int a = ( arr[ 2 ] + arr[ 1 ] );`
`int b = ( arr[ 3 + 1 ] + 20 );`	

19. Given that **int x[ ] [ ] = {{2, 4, 6}, {3, 5, 7}};** what will be the value of **x[1][0]** and **x[0][2]** ? **ICSE 2009**	
20. State the total *size* in bytes, of the array **a[4]** of ***char*** data type and **p[4]** of ***float*** data type. **ICSE 2011**	
21. Write statement to find the *length* of a *character array* named **ch**. **ICSE 2013**	

22. If **int[ ] n ={1, 2, 3, 5, 7, 9, 13, 16}**; what are the values of x and y?

```
x =    Math.pow( n[ 4 ], n[ 2 ] );
y =    Math.sqrt( n[ 5 ] + n[ 7 ] );
```

ICSE 2013

23. Find errors in the given program code and re-write the statements correctly to assign values to an integer array.

```
    int a = new int( 5 );
for( int i = 0; i <= 5; i++ )  a[ i ] = i;
```

ICSE 2014

24. What will the following code print?

```
    int arr[ ] = new int[ 5 ];
System.out.println( arr );
```

ICSE 2015

(i) 0 (ii) value stored in arr[0]

(iii) 0000 (iv) garbage value

25. If **int x[ ] = { 4, 3, 7, 8, 9, 10 };** what are the values of ***p*** and ***q***?

(i) p = x. length

(ii) q = x[2] + x[5] * x[1];

ICSE 2016

26. Perform the following operations:

(i) Declare a two dimensional integer array **m** of **3 rows** and **3 columns**.

(ii) Print the length of array **x[ ][ ]**.

(iii) Initialize a two dimensional array **m[ ][ ]** of **2 rows** & **3 columns** with fractional values.

(iv) Assume an array **z[4][5]** with integers. Write print statements to display elements of **1st row** and **3rd column**, and print the very first integer of the array.

27. Assume the array **m[ ][ ] = {{3, 5, 1}, {5, 6, 7}};**

Give the output of the following statements:

(i) System.out.println(m[1][1] * 10);

(ii) System.out.println(m . length);

(iii) System.out.println(m[1] . length);

(iv) System.out.print(m[0][2] + " " + m[1][2]);

28. Assume the array **A[ ][ ] = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};**

Give the output of the following statements:

(i) System.out.println(A[2][1]* A[1][2]);

(ii) System.out.println(A . length + 1);

(iii) System.out.println(A[2].length + A[0].length);

(iv) System.out.println(A[2-1][1*2]);

29. Initialize a two dimensional array **ar[ ][ ]** having only even integers from 1 to 10 in *2 rows* and *2 columns*.

Questions	Space provided for writing output

1. Given the following function to perform a task. Write a function **show( )** that includes everything necessary to invoke the given function.
`int process ( int y )`
`{`
`return y * 3;`
`}`	

2. Consider the following class and answer the questions that follow:
`class someFn`
`{`
`int n;`
`void process( )`
`{`
`int n;`
`n = 25;`
`} }`
(i) State whether the value 25 is assigned to local variable '*n*' or an instance variable '*n*'.
(ii) Can **n=25** be used in other method of class?	

3. What will be the output of following when function **outFinal( )** is invoked?
`class process`
`{`
`int ar[ ]={ 2, 5, 8, 9, 1, 6, 4, 9, 3, 5 };`	

```
int result = 0;
public void Display( )
{
    for( int i = 0; i < 12; i += 2 )
    result + = ar[ i ];
    System.out.println("Output = " + result );
}
public void outFinal( )
{
    Display( );
}
}
```

4. Give the output when function **result1( )** is invoked. State the reasons for each output.

```
public class  theoutput
{
int  num=10;
public  void result1( )
{
    System.out.println( num );
int  num =25;
    result2( num );
System.out.println( num );
    result3( );
System.out.println( num );
}
public void  result2( int num )
{
    num = 30;
    System.out.println( num );
}
public  void  result3( )
{
    System.out.println( num );
    num = 40;
}
}
```

5. What will be the output of following method for?

(i) fun(28,39)

(ii) fun(27,39)

(iii) What the given method is computing? State in one line.

```
public boolean fun( int a, int b )
{
// assume a, b are positive
boolean  c = false;
```

```
while( a>1 && b > 1)
{
  if( a > b)  a -= b;
          else   b -= a;
}
if( ( a == 1) || ( b == 1 ) )
  c = true;
return ( c );
}
```

6. Identify the error if any in the following part of the program code, correct it and rewrite the program code with no error:

```
public  class  function( )
{    int  a;
public  void  display( int  x )
{
     a = x
System.out.println("\n a = " + a + " x = " + x );
}
public  void  invoke( );
{
  this.display( );
}
}
```

7. Read following program code and answer the questions that follows:

```
class  numberProcess  .
{
public  static int  x;
public static void  process( numberProcess   num )
{
num . x ++ ;
}
}
class  result
{
public static  void  show( )
{
                                     // line number 1
numberProcess    obj = new numberProcess( );

obj . x = 200;                       // line number 2
obj.process( obj );                  // line number 3
System.out.println( obj . x);
}
}
```

(i) Explain the statement shown as line number 1.

(ii) Explain the statement shown as line number 2.

(iii) Explain the statement shown as line number 3.

(iv) What is the output of the above program code?

8. Give the output when method **print( )** runs:

```
class  text
{
public void demo( String nam )
{
    nam = "good";
}
public void print( )
{
    String na = "good";
System.out.println( na );
    demo( na );
System.out.println( na );
} }
```

9. Consider the following program coding and answer the questions that follow:

```
class  someprocess
{
float  area( float  radius )
{
    return 3.14* radius * radius;
}
float  area( float length, float breadth )
{
    return length * breadth ;
}
float area ( float a, float b, float c )
{
float  s = ( a + b + c )/2;
float arr = Math.sqrt( s *(s – a)*(s – b)*(s – c));
return arr;
}
}
```

(i) Which of the OOP'S concept is implemented by the above class?

(ii) What is being computed by all the functions of the class?

10. Give the output of following function when invoked:

```
public static void func( )
{    long  num = 432432;
num = num / 1000;      long n = num / 100;
System.out.println( num+ "\t" + n);
}
```

11. Give the output of following program code when function **output( )** is invoked?

```
class testOutput
{
public testOutput( )
{
    Display( );
    System.out.print("Constructor");
}
public void Display( )
{
    Printing( );
    System.out.print(" use of" );
}
public void Printing( )
{
    System.out.print(" this is the");
}
public static void output( )
{
testOutput obj = new testOutput( );
}
}
```

12. Read the following function carefully and answer the questions that follows:

```
public int SomeFunc( int m, int b)
{
int out = 1;
for( int x = 1; x <= b; x++)
{
    out *= m;
}
return ( out );
}
```

(i) Give the return value of variable *out*, if **m=3, b=2**?
(ii) Give the return value of variable *out*, if **m=5, b=3**?
(iii) What activity is performed by the above function? State in one line.
(iv) Name the pre-defined function (if any) which could perform the same process as defined in the above function.

13. Give the prototype of a function ***show( )*** which receives string ***ch*** and an integer ***h*** and returns *a* ***character***.

14. Give the prototype of a function ***check( )*** which receives character ***ch*** and an integer ***n*** and returns ***true*** or ***false***.
ICSE 2010

15. Give the prototype of a function ***search( )*** which receives a sentence in ***sentnc*** and a word in ***wrd*** and returns **1** or **0** ? **ICSE 2011**	
16. Name the Java keyword that : (i) Indicates a method has no return type. (ii) Stores the address of the currently calling object. **ICSE 2013**	
17. State, whether the following statement is a valid method invocation or not: System.out.println("Java"); **ICSE 2014**	

18. What are the values of ***a*** and ***b*** after the following function is executed, if the value passed are **30** and **50** ?
void paws(int a, int b)
{ a = a + b;
b = a - b;
a = a - b;
System.out.println(a + "," + b); } **ICSE 2015**	

19. Write a function prototype of the following: A function ***PosChar*** which takes a ***string*** *argument* and a ***character*** *argument* and returns an ***integer***. **ICSE 2016**	
20. What are the two ways of function invoking? **ICSE 2017**	

Character Functions, String Handling and Error Handling

Questions	Space provided for writing output
1. Give output of following statements: (i) System.out.println("Lucknow".toUpperCase( )); (ii) System.out.println("England".charAt(4)); (iii) System.out.println("Nice".equals("nice")); (iv) String st = "this is final examination"; System.out.println(st.length( ));	
2. Give the output of the following statements : (i) System.out.print("abc". indexOf('b')); (ii) System.out.print("S".toLowerCase( )); (iii) System.out.print("India".compareTo("country"));	
3. What will be output of the following statements? String a = "Computer", b = "Applications";	

(i) System.out.println(a + b);
(ii) System.out.println(a . substring(3));
(iii) System.out.println(a . equals(b));
(iv) System.out.println(a . charAt(2));

4. Write valid java statements to perform the following tasks on strings:
   (i) Extract first 10 characters from a string object **str**.
   (ii) To print the position of the first occurrence of the letter '**B**' in the string object **str**.
   (iii) Print the length of the string stored in object **str**.
   (iv) Convert the string stored in **str** in uppercase form.

5. Give output of the following :
   (i) System.out.println("ComPUter".toUpperCase( ));
   (ii) System.out.println("ProGRam".charAt(4));
   (iii) System.out.println("My Book".length( ));
   (iv) System.out.println("great" . indexOf('e'));
   (v) System.out.println("great".equalsIgnoreCase ("GREAT"));
   (vi) String ss = "Good".concat(" Bye"); System.out.println(ss);

6. Write a single line statement to implement the following if:
   String text[ ] = {"Can", "Don", "Block", "Man", "Bye", "Cos", "Sine", "Good", "Yes"};
   (i) Count the number of string in the object **text**.
   (ii) Find the number of characters of the strings at 1st and 3rd indexes.
   (iii) Print the first character of 4th index string from the array.
   (iv) Write a statement to check string stored at 0th and 5th indexes are same or not without cases.

7. Write valid Java statements to perform following tasks on strings:
   (i) Initialize a string object **city** to store "KOLKATA".
   (ii) Print the number of characters present in the string object **city**.
   (iii) Concatenate "MUMBAI" to the object **city** and store the output in the object **result**.
   (iv) Extract and print character present at **4th** and **7th** indexes from the object **result**.
   (v) Compare the object **city** with "SINGAPORE" and store the boolean result in the variable **res**.
   (vi) Compare the object **city** with "SINGAPORE" and store the output in the integer variable **R**.

Output Questions

8. Read the given array of strings answer the questions:

String st[ ] = {"SINGAPORE", "MUMBAI", "DELHI", "BANGLORE"};

(i) What is the range of indices of the array **st** ?

(ii) What is size of the array **st** ?

(iii) What are the results of **st[3].length( );** and **st[0].length( );** ?

(iv) What is the result of **st[1].toLowerCase( );** ?

(v) What is the result of **st[2].reverse( );** ?

(vi) What is the result of **"MUMBAI".replace('M', 'H');** ?

9. Give output of following program code:

```
String str = "COMPUTER";
char ch = str.charAt( 9 / 2 );
int as = ch;
System.out.println( char( as ));
```

10. Write a valid Java statement to assign 8 names of countries in the string array.

11. Give output of following program code if method is invoked as **test("XYZ");?**

```
void test( String n )
{
String st = n + "ABC";
System.out.println( "n = " + n );
System.out.println( "st = " + st );
}
```

12. What do the following functions **return** for?

String x = "hello"; String y = "world";

(i) System.out.println(x + y);

(ii) System.out.println(x . length( ));

(iii) System.out.println(x . charAt(3));

(iv) System.out.println(x . equals(y));

(v) System.out.println(x.equalsIgnoreCase(y));

13. Consider the following program code. Underline the errors and rewrite the correct program code:

```
Public Class test
{
public static Void main( String args[ ] )
{
String ns = "10th std ;
System.out.println("I am class X student");
System.out.println("I am studying in" ns" + "\t");
system.out.pritln("This is the output);
} }
```

14. Write a statement to declare and initialize a character array of first five lowercase consonants.	
15. Give the output for following program code: String sat = "ExAmInATioN"; for(int y=0; y < sat.length( ); y += 2) System.out.print(sat.charAt(y) + " ");	

16. Underline the **errors** in following program code and write the correct along with output:
String "Sri Lanka" = str;
StringBuffer st = str.LowerCase( );
for(int x =0; x < st.length ; x += 2)
{
system.out.println(st.charAt(x));
}	

17. State the output of following function when invoked?
public void show( )
{ String val = "examinations";
for(int i = 0; i < 12; i ++)
{
for(int j = 1 ; j < 2 ; j++)
{
System.out.print(val.charAt(i) + "\t");
}
} }	

18. Give the output of following function when invoked?
public void show( )
{ String str = "APPS";
for(int i = 0; i < str.length( ); i ++)
{
for(int j = 0 ; j <= i ; j++)
{
System.out.print(str.charAt(j));
}
System.out.print("\n");
} }	

19. State the output of following function when invoked:
public void show( )
{
String str = "COLLEGE";
for(int i = 0; i < str.length( ); i ++)
{
for(int j = 0 ; j <= i ; j++)
{	

```
String text = "UPPER";
for(int k = text.length( )-1; k >= 0; k--)
{
for( int y = 0; y <= k; y++ )
{
System.out.print( text.charAt( y ));
}
System.out.print("\n"); }
```

21. Consider the following function :
`public void show (String S)`
`{`
`if (S. length( ) > 10)`
`System.out.println(S.substring(11,S.length() ));`
`else`
`System.out.println(S.toLowerCase( ));`
`}`
(i) What is the output of the above function if the input to the function is; "**This is Program**"?
(ii) What is the output of the above function if the input to the function is : "**CAPTAIN**" ?	

22. Give the output of the following code :
`String A = "26", B = "100";`
`String D = A + B + "200";`
`int x = Integer.parseInt (A);`
`int y = Integer.parseInt (B);`
`int d = x + y + 200;`
`System.out.println ("Result1 = " + D);`
`System.out.println ("Result2 = " + d);`	

23. Give the output of the following code :
`long x = 30, y = 40;`
`String A, B; String S;`
`A = String.valueOf (x);`
`B = String.valueOf (y);`
`System.out.println("Output1 = " + x + y);`
`System.out.println("Output2 = " + A + B);`
`S = A + B;`
`System.out.println("Output3 = " + (x + y));`
`System.out.println("Output 4 = " + S);`	

24. Give the output of the following code :
`System.out.println("Finder".endsWith("der"));`
`System.out.println("Better".startsWith("Bet"));`	

25. Write valid Java statements for following: (i) To input a name into string object ***text*** from key-board. (ii) To find and print the total number of characters present in the object ***text***. (iii) To input a name into object ***nam*** and perform various comparisons on ***text*** and **nam**. (iv) To convert ***text*** into upper case. (v) Extract 2nd last character of a word stored in the variable ***wd***. **ICSE 2010**	
26. Write a statement of cheek if the second character of a string ***str*** is in uppercase ? **ICSE 2010**	
27. Give the output of the following : String n = "Computer Knowledge"; String m = "Computer Applications"; System.out.println(n.substring(0, 8). concat(m.substring(9))); System.out.println(n . endsWith("e")); **ICSE 2011**	
28. Write the output of the following : (i) System.out.println(Character.toUpperCase('R')); (ii) System.out.println(Character.toUpperCase('j')); **ICSE 2011**	
29. State the output of the following program segment: String s = "Examination"; int n = s.length( ); System.out.println(s.startsWith(s.substring(5, n))); System.out.println(s. charAt(2)== s. charAt(6)); **ICSE 2012**	
30. State the method that determines, if a specific character is an uppercase character? **ICSE 2012**	

31. State the data type and values of ***a*** and ***b*** after the following segment is executed.
String s1 = "Computer" , s2 = "Application";
a = (s1.compareTo(s2));
b = (s1.equals(s2));
System.out.println(s1.charAt(2)==s2.charAt(6));
ICSE 2012	

32. What will the following code output? String s = "malayalam"; System.out.println(s.indexOf('m')); System.out.println(s.lastIndexOf('m')); **ICSE 2012**	
33. State the values stored in string variables **str1** & **str2**. String s1 = "good"; String s2 = "world matters"; String str1 = s2.substring(5).replace('t', 'n'); String str2 = s1.concat(str1); **ICSE 2013**	

Output Questi...

34. State the output of the following program segment: String str1 = "great"; String str2 = "minds"; System.out.println(strl.substring(0, 2).concat(str2.substring(1))); System.out.println(("WH" + (strl.substring(2).toUpperCase( )))); **ICSE 2014**	

35. State the value of **characteristic** and **mantissa** when the following code is executed.
String s = "4.3756"; int n = s.indexOf('.');
int characteristic=Integer.parseInt(s.substring(0,n));
int mantissa = Integer.valueOf(s.substring(n+1));
ICSE 2014	

36. What will be the output when the following code segments are executed?
(i) String s = "1001";
int x = Integer.valueOf(s);
double y = Double.valueOf(s);
System.out.println("x = " +x);
System.out.println("y = " +y);
(ii) System.out.println("The King said \"Begin at the beginning!\" to me.");
ICSE 2014	

37. What is the value stored in **res** given below: double res = Math.pow("345".indexOf('5'), 3); **ICSE 2015**	
38. State the data type and value of **y** after the given statements are executed : char x = '7'; y = Character.isLetter(x); **ICSE 2015**	
39. State the output of the following program code: String a = "smartphone", b = "Graphic Art"; String h = a . substring(2, 5); String k = b . substring(8) . toUpperCase( ); System.out.println(h); System.out.println(k . equalsIgnoreCase(h)); **ICSE 2015**	
40. Name the string function which removes the blank spaces provided in the prefix and suffix of a string. **ICSE 2015**	
41. Give the output of the following ***string*** functions: (i) "MISSISSIPPI" . indexOf('S') + "MISSISSIPPI" . lastIndexOf('I') (ii) "CABLE".compareTo("CADET"); **ICSE 2016**	
42. Write the ***return type*** of the following library functions: (i) isLetterOrDigit(char) (ii) replace(char, char); **ICSE 2016**	

43. Name the keyword that informs that an error has occurred in an input/output operations. **ICSE 2011**

44. Name the type of error (syntax, runtime or logical error) in each case given below:
(i) Division by a variable that contains a value zero.
(ii) The multiplication operator used when the operation should be division.
(ii) Missing semicolon. **ICSE 2012**

45. Name the type of error (syntax, runtime or logical error) in each case given below:
(i) **Math.sqrt(36-45)**
(ii) **int a; b; c;** **ICSE 2016**

46. Consider the following code and answer the questions that follow : **ICSE 2009**

```
class  academic
{
      int x, y;
      void  access( )
      {
            int a, b;
            academic student = new academic( );
      System.out.println ("object created");
      }
}
```

(i) What is the object name of class **academic** ?
(ii) Name the class variables used in the program.
(iii) Write the local variables used in the program.
(iv) Give the type of function used and its name.

47. Name the error-handler that returns the user defined messages instead of the standard error message while an error has occurred in a program code.

48. Name the package used with error handler **throws IOException**.

49. Name the type of error for the following statement:
System.out.println(arr[-2]);

50. Name the type of error (syntax, runtime or logical error) in each case given below:
(i) Math.sqrt(-49)
(ii) String na, int x;
(iii) for(int y = 10; y > 0; y++)

51. State the *type of data* and value of **res** after the following is executed: **ICSE 2017**

```
char  ch = 't';
res = Character . toUpperCase( ch );
```

Output Questions

52. Write the output of the following code: **ICSE 2017**

```
char ch = 'F';
int m = ch;                m = m + 5;
System.out.println( m + " " + ch);
```

53. Give the output of the following: **ICSE 2017**

```
String  s = "Today is Test";
System.out.println( s.indexOf( 'T' ));
System.out.println( s.substring(0, 7) + " " + "Holiday");
```

54. Give the output of the following : **ICSE 2017**

```
String  A = "26", B = "100";
String  D = A + B + "200";
int x = Integer.parseInt( A );
int y = Integer.parseInt( B );
int d = x + y;
System.out.println("Result 1 = " + D);
System.out.println("Result 2 = " + d);
```

55. Give the output of the following : **ICSE 2017**

```
String  x [ ] = { "SAMSUNG", "NOKIA", "SONY",
"MICROMAX", "BLACKBERRY"};
```

(i) `System.out.println( x[ 1 ] );`

(ii) `System.out.println( x[ 3 ] . length( ) );`